Analog quantum devices for optimization problems

DuQuantum 2026: Pasqal's Challenge

Team Name: Neutral Groundstate

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1 Problem Statement

The **Capacity Constrained Clustering problem** asks: given a depot, N customers with known demands d_j, and K identical vehicles each of capacity Q, assign every customer to exactly one vehicle (never exceeding Q), equivalently one cluster.

2 QUBO Formulation with Slack Variables

We formulate the the capacity-constrained clustering problem as an instance of the Quadratic Unconstrained Optimization Problem (QUBO).

Let the k in the range $[1, K]$ represent the vehicle number.

Let Q_k be the capacity per $k-$th vehicle. Since all vehicles are identical, the capacity value will be the same for all vehicles.

Let the j in the range $[1, N]$ represent the customer, where N total number of customers.

Let d_j be the demand for each $j-$th customer.

We assume that one vehicle must visit one cluster only, therefore saying $k-$th vehicle, means choosing $k-$th cluster.

Let d_{ij} be the distance between customer $i-$th and customer $j-$th. The matrix D encodes pairwise distances on its (i, j) entry.

Let x_{jk} represent an assignment of the $j-$th customer to $k-$th vehicle. We denote by X the vector of customers-to-vehicles assignments. Formally, X represents a vector of the **binary decision variables**.

$$\begin{aligned} \text{optimize} \quad & x^T D x \\ \text{subject to} \quad & \sum_{j=1}^{N} d_j \leq Q_k \quad \forall k = 1, \ldots, K, \\ & \sum_{k=1}^{k} x_{jk} = 1 \quad \forall j = 1, \ldots, N \\ & x \in \{0, 1\}. \end{aligned} \tag{1}$$

We reformulate the problem in 1 as a QUBO

$$\begin{aligned} \text{optimize} \quad & H \\ & H = \frac{1}{2} X^T D X + \alpha (Q_k - \sum_{j=1}^{N} d_j - S_j)^2 + \beta (\sum_{k=1}^{k} x_{ij} - 1)^2, \end{aligned} \tag{2}$$

where we can express a slack variable in its binary representation $S_j = \sum_{M}^{r=0} 2^r s_r$ for $M = \lfloor \log_2 (Q_k - \sum_{j=1}^{N} d_j + 1)) \rfloor$.

2.1 QUBO Formulation with Unbalanced Penalization

As the number of slack variables increases with the increase in the number of constraints, we consider formulation of the problem in an unbalanced penalization style[1], where one does not have to rely on the slack variables but rather on a thoughtfully chosen penalization parameters of the QUBO formulation.

As such we formulate a slack-free optimization model as follows. We denote the capacity constraints as a separate function

$$h(x) = Q_k - \sum_{j=1}^{N} d_j \geq 0. \tag{3}$$

We then formulate the penalization from the Taylor expansion of the $exp(-h(x)) \approx -h(x) + \frac{h^2(x)}{2}$. Meaning that we add a term $\zeta(x) = -\lambda_1 h(x) + \lambda_2 h(x)$, where the tuning parameters λ_1 and λ_2 must be chosen carefully in order to not over penalize and not to under penalize.

optimize: H

$$H = \frac{1}{2}X^T DX + \beta(\sum_{k=1}^{K} x_j k - 1)^2 - \lambda_1(Q_k - \sum_{j=1}^{N} d_j) + \lambda_2(Q_k - \sum_{j=1}^{N} d_j)^2, \tag{4}$$

2.2 QUBO Formulation with Slack Variable Reduction

One other possibility to reduce the number of slack variables we explored is the Variable Reduction Technique [2]. The techniques uses the binary slack variables by normalizing inequality constraints as follows. For an inequality constrain

$$\sum_{j=1}^{N} d_j \leq Q_k \tag{5}$$

consider to normalize by Q_k

$$\sum_{j=1}^{N} \frac{d_j}{Q_k} \leq 1. \tag{6}$$

Then we consider the QUBO formulation as follows.

optimize: H

$$H = \frac{1}{2}X^T DX + \alpha(1 - \sum_{j=1}^{N} \frac{d_j}{Q_k} - 1 - s_j)^2 + \beta(\sum_{k=1}^{K} x_i j - 1)^2, \tag{7}$$

where $s_j \in 0, 1$ is a binary slack variable.

3 Hardware and Pulse Design

A Pasqal neutral-atom device evolves under the Hamiltonian

$$\frac{H(t)}{\hbar} = \frac{\Omega(t)}{2} \sum_i \hat{\sigma}_i^x - \delta(t) \sum_i \hat{n}_i + \sum_{i<j} \frac{C_6}{\|\mathbf{r}_i - \mathbf{r}_j\|^6} \hat{n}_i \hat{n}_j, \tag{8}$$

where $\Omega(t)$ (Rabi frequency) drives exploration, $\delta(t)$ (detuning) encodes linear QUBO coefficients, and the C_6/r^6 interaction encodes quadratic couplings through atom positions. Each atom represents one binary variable x_{jk}.

The mapping between the QUBO formulation as given in 2 and the Hamiltonian in 8 can be accomplished by expanding a QUBO formulation and collecting quadratic and linear terms in either 1, or 4, or 7 to map the quadratic term $\sum_{i<j} \frac{C_6}{\|\mathbf{r}_i - \mathbf{r}_j\|^6} \hat{n}_i \hat{n}_j$ in 8 and the linear terms to map the $-\delta(t) \sum_i \hat{n}_i$ on the device Hamiltonian in 8.

We followed the hardware restrictions and the chose of hardware parameters as given in the *Pasqal Challenge* document specification.

4 Results

For the distance we used a pairwise distance between customers.

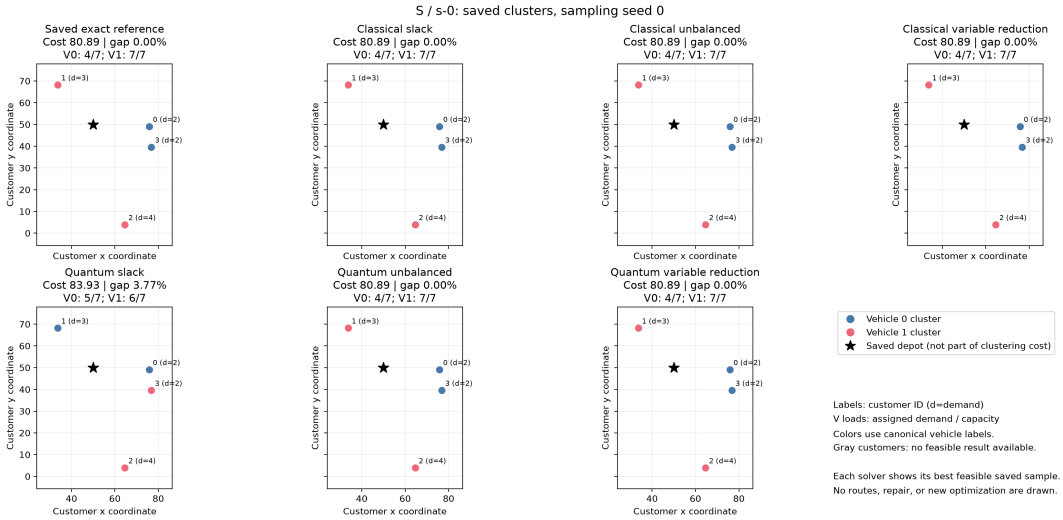


Figure 1: Comparison of classical and quantum solvers of the clustering problem for a sample instance of S-Tier Problem with the demand $d_j = (2, 3, 4, 2)$. The tuning parameters for the Unbalanced Penalization were $\lambda_1 = 0.2$, $\lambda_2 = 0.01$.

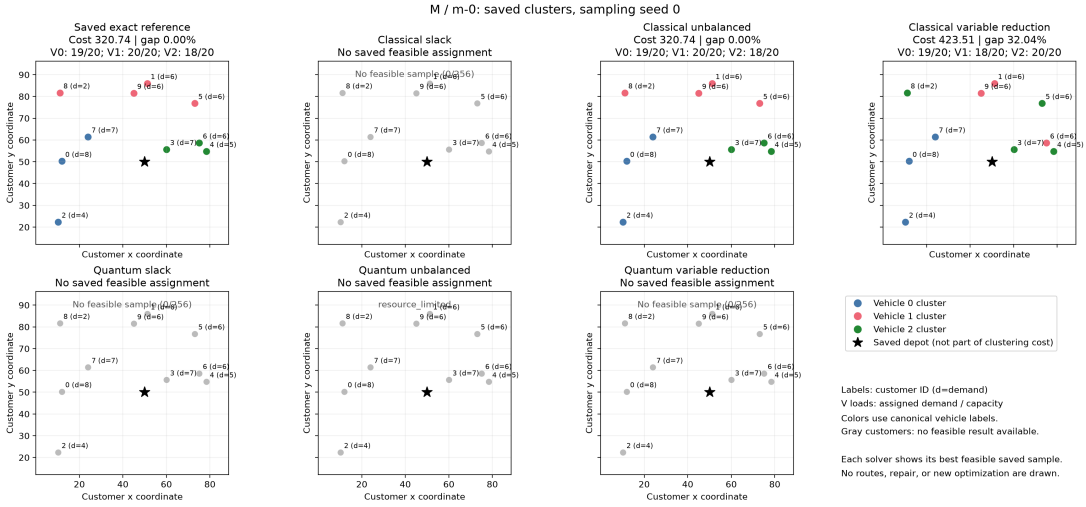


Figure 2: Comparison of classical and quantum solvers of the clustering problem for a sample instance of M-Tier Problem with the demand $d_j = (8, 6, 4, 7, 5, 6, 6, 7, 2, 6)$. The tuning parameters for the Unbalanced Penalization were $\lambda_1 = 0.75$, $\lambda_2 = 0.02$.

Encoding	Tier S	Tier M	Tier L	≤80?
Assignment only ($N \times K$)	8	30	60	✓
+ linear slack ($K \times Q$)	22	90	180	×
+ log slack ($K\lceil \log_2 Q \rceil$)	14	45	80	barely
Unbalanced penalisation (ours)	8	30	60	✓

Table 1: Theoretical variable counts per tier. Unbalanced penalisation eliminates slack entirely.

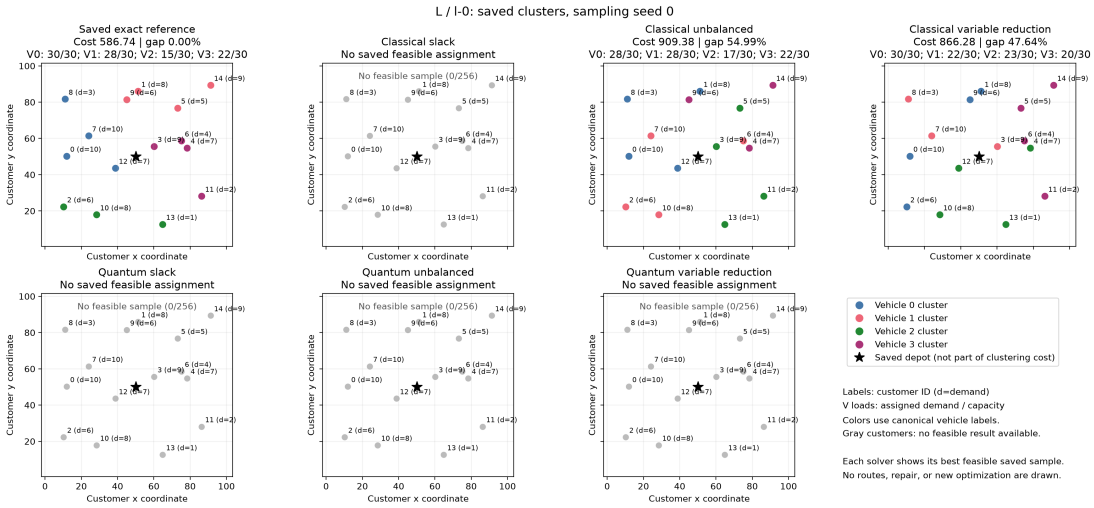


Figure 3: Comparison of classical and quantum solvers of the clustering problem for a sample instance of L-Tier Problem with the demand $d_j = (10, 8, 6, 9, 7, 5, 4, 10, 3, 6, 8, 2, 7, 1, 9)$. The tuning parameters for the Unbalanced Penalization were $\lambda_1 = 0.75$, $\lambda_2 = 0.02$

5 Challenges and Questions Unanswered

We observed that some models can not produce a feasible bitstring solution for the M and L Tier 2 3. There are several issues to investigate in here including a different range of QUBO penalization parameters, different chose of the coupling strength for the device Hamiltonian parameter, different grid shape.

The unbalanced penalization formulation for the M and L tier could show a larger gap on Tier M, suggesting the penalty weights λ_1, λ_2 require instance-specific tuning beyond our 24-hour budget. The larger gap may not necessarily be a signal of a poorly performed model. A larger gap is only indicating a difference in energy level between classical version of a Hamiltonian and the quantum Hamiltonian. However, as soon as the both minimum produce the same valid bitstring, we may claim that model if well performing and producing reliable results.

One of the grand challenge we identify was in meticulous and thoughtful investigation of different parameters of our QUBO model once we verified a base model prototyped.

AI usage. Microsoft Copilot was used for code generation, and debugging, and Claude (Anthropic) was used throughout for code literature guidance and LaTeX formatting. All derivations, penalty choices, and results were reviewed and owned by the team.

Tier	Load ratio	Largest-demand ratio
S	0.785714	0.571429
M	0.950000	0.400000
L	0.791667	0.333333

Table 2: Dimensionless demand ratios by tier.

Encoding (QUBO formulation + solver family)	Variable-count formula (bits)	S tier bits / quantum atoms	M tier bits / quantum atoms	L tier bits / quantum atoms	Fits in 80 atoms? All displayed tiers ≤ 80
Classical slack	$NK + K\lceil \log_2(C+1) \rceil$	14	45	80	Yes
Classical unbalanced	NK	8	30	60	Yes
Classical variable reduction	$NK + K$	10	33	64	Yes
Quantum slack	$NK + K\lceil \log_2(C+1) \rceil$	14	45	80	Yes
Quantum unbalanced	NK	8	30	60	Yes
Quantum variable reduction	$NK + K$	10	33	64	Yes

Table 3: QUBO variable counts by encoding and solver family. N is the customer count, K the vehicle count, and C the capacity per vehicle. Counts are binary variables (bits); the quantum mapping requires one atom per bit. The 80-atom check covers all displayed tiers and checks count only, not geometry or mapping fidelity.

References

[1] J. A. Montañez-Barrera, Dennis Willsch, A. Maldonado-Romo, and Kristel Michielsen. Unbalanced penalization: a new approach to encode inequality constraints of combinatorial problems for quantum optimization algorithms. *Quantum Science and Technology*, 9(2):025022, 2024.

[2] Amit Verma and Mark Lewis. Variable reduction for quadratic unconstrained binary optimization, 2021.